Henry Smith

Sequoia Hall, Stanford, CA, 94305
smithhd@stanford.edu | https://smithhenryd.github.io/ | github.com/smithhenryd

Overview

PhD candidate in Statistics at Stanford (Knight–Hennessy Scholar, NSF Fellow) interested in probabilistic generative modeling (e.g., diffusion, flow models) and scalable Bayesian inference for stochastic dynamical systems. My research develops models and inference algorithms for scientific data, with applications to neuroscience and computational protein design.

Education

Stanford University, Palo Alto, CA 2023 –
Ph.D., Statistics
Advisors: Scott Linderman and Brian Trippe
Relevant Coursework: stochastic processes, probability theory, theoretical statistics, information theory

Stanford University, Palo Alto, CA 2023 – 2025
M.S., Statistics

Yale University, New Haven, CT 2018 – 2022
B.S., Statistics & Data Science, Mathematics
summa cum laude
Thesis: "Implicit Regularization in Deep Learning: The Kernel and Rich Regimes"
Advisor: Harrison Zhou

Work Experience

Research Intern Jun 2026 – Aug 2026
Flatiron Institute, Simons Foundation *New York City, NY*
Advisors: Minhuan Li, Mark Goldstein, Pilar Cossio

- Studied biases in machine learning algorithms that reconstruct 3D molecular structures from noisy cryo-electron microscopy (cryo-EM) images; proposed an alternative metric that corrects for these biases
- Collaboration ongoing; working toward publication

Research Assistant (full-time) Oct 2022 – Oct 2023
University of Cambridge, Department of Computer Science & Technology *Cambridge, UK*
Advisor: Amanda Prorok

- Mathematically formalized and programmed geometric deep learning algorithms to enable close proximity flight of aerial drones (e.g., for search and rescue tasks); validated in simulation and real-world experiments

Student Research Assistant Jun 2021 – Jun 2023
Cornell University, Department of Mathematics *Ithaca, NY*
Advisors: Alex Townsend, Nicolas Boullé

- Developed mathematically interpretable deep learning algorithms for the unsupervised discovery of linear partial differential equations (PDEs)

Awards & Honors

Departmental Teaching Award[1] | *Stanford Department of Statistics*	Jun 2025
Knight–Hennessy Scholarship[2] | *Stanford University* | $300,000	May 2024
Graduate Research Fellowship | *National Science Foundation (NSF)* | $147,000	Sep 2023
EDGE (Enhancing Diversity in Graduate Education) Fellowship | *Stanford University* | $12,800	Sep 2023
Outstanding Senior Thesis Award in Statistics & Data Science[3] | *Yale University*	May 2022
Emerson Tuttle Cup[4] | *Yale University*	May 2022
Phi Beta Kappa | *Yale University*	Feb 2022
Research Experience for Undergraduates (REU) | *National Science Foundation (NSF)* | $5,000	Jun 2021
Google Summer of Code Fellowship | *Google* | $6,000	May 2020
Richter Summer Research Fellowship | *Yale University* | $1,000	May 2020

Publications

[1] **H.D. Smith**, S.W. Linderman, B.L. Trippe. Closing the approximation gap in simulation-free latent SDEs. *Advances in Neural Information Processing Systems* (NeurIPS), 2026.

[2] H. Lee, **H.D. Smith**, C. Tang, A. Hu, X. Sun, D. Anderson, S.W. Linderman. Gaussian process-based dynamical systems models for unsupervised neural dynamics discovery. In submission, 2026.

[3] J. Lunger, **H.D. Smith**, S.W. Linderman. Learning biomechanistic models of whole-brain dynamics with memory efficient natural gradient variational inference. In submission, 2026.

[4] **H.D. Smith**, N.L. Diamant, B.L. Trippe. Calibrating generative models to distributional constraints. *International Conference on Machine Learning* (ICML), 2026.

[5] H. Lee, **H.D. Smith**, A. Hu, S.W. Linderman. Modeling shared and individual dynamics with hierarchical Gaussian process dynamical systems. *Computational and Systems Neuroscience* (COSYNE), 2025.

[6] A. Hu*, **H.D. Smith***, S.W. Linderman. SING: SDE inference via natural gradients. *Advances in Neural Information Processing Systems* (NeurIPS), 2025.

[7] **H.D. Smith**, B.L. Trippe. Calibrating generative models. International Conference on Machine Learning (ICML) Generative AI and Biology Workshop, 2025. [Spotlight] (10% of accepted submissions selected for talk)

[8] **H.D. Smith**, A. Shankar, J. Gielis, J. Blumenkamp, A. Prorok. SO(2)-equivariant downwash models for close proximity flight, *Robotics and Automation Letters* (RA-L), IEEE, 2024.

∗ denotes equal contribution

[1] Remarks from students and faculty: "extremely conscientious, proactive, and well prepared"; "the strongest TA in 10 years"

[2] Full-ride scholarship for graduate studies at Stanford; scholars selected on the basis of independence of thought, purposeful leadership, and civic mindset; 89 scholars selected from ∼8000 applicants

[3] One of two graduating seniors selected to receive the award by the Statistics & Data Science department faculty

[4] Presented annually at Commencement to the Davenport College senior most distinguished for scholastic attainments; there were approximately 130 students in the graduating class

Teaching

Teaching Assistant:

STATS 200: Introduction to Theoretical Statistics | *Stanford University*	Autumn 2026
STATS 315: Modern Applied Statistics, Learning | *Stanford University*	Winter 2026
STATS 202: Statistical Learning and Data Science | *Stanford University*	Fall 2024
STATS 60: Introduction to Statistics | *Stanford University*	Spring 2024
S&DS 365/665: Intermediate Machine Learning | *Yale University*	Spring 2022
S&DS 262/562: Computational Tools for Data Science | *Yale University*	Spring 2021
S&DS 100/500: Introduction to Statistics | *Yale University*	Fall 2019

Service

Reviewer, *Advances in Neural Information Processing Systems* (NeurIPS)	2026
Reviewer, *International Conference on Machine Learning* (ICML)	2026
Stanford Statistics PhD Admissions Committee | Student Member	2026
Stanford Women and Allies in Statistics | Member, Event Organizer	2023 –
Reviewer, *International Conference on Learning Representations* (ICLR)	2025
Stanford Department of Statistics Qualifying Exam Reader	2025
Reviewer, *Learning for Dynamics and Control Conference* (L4DC)	2025

Technical Skills

Languages: Python, R, C, MATLAB, SQL
Frameworks: JAX, PyTorch
Research tools: Git, Linux/HPC, Docker, SLURM
Open-source contributions

- `sing`: https://github.com/lindermanlab/sing
- `cgm`: https://github.com/smithhenryd/cgm
- `helmholtz-sde`: https://github.com/lindermanlab/helmholtz-sde